

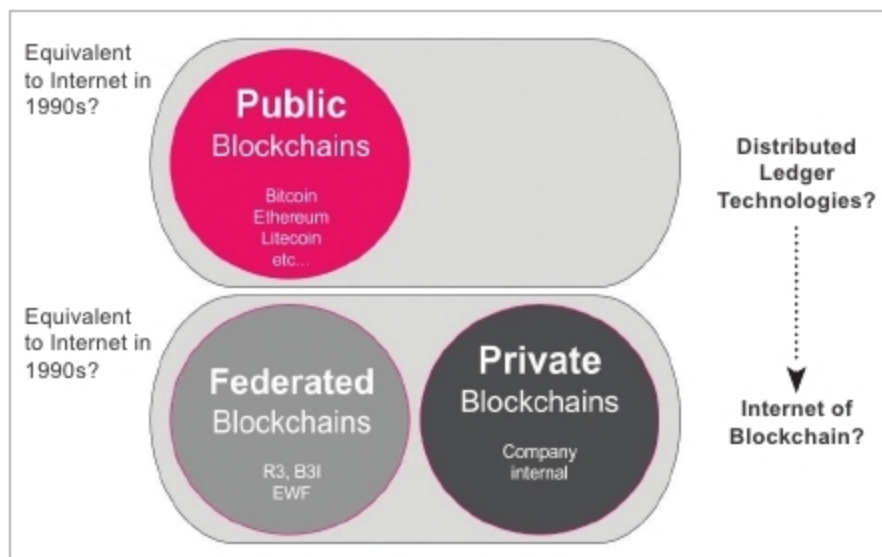


Figure 3: Types of blockchains (Image source: [googleimages.com](https://www.googleimages.com))

Private blockchain: Private blockchains contain different groups and participants who can easily verify various transactions internally. Here, write permissions are kept centralised within just one organisation whereas read permissions might be public or restricted to a certain extent. For instance, database management and auditing applications are always internal to a single company, so public readability may not be necessary at all, in certain cases. In some cases, public audit permissions may be desired. Private blockchains are really valuable for their efficiency and security against fraud within various financial institutions. For example, MONAX, Multichain, etc, are private blockchains.

Listed below are some of the best open source blockchain platforms.

Ethereum

This is an open source blockchain platform that helps in running smart contracts and provides different programming tools to create them. It was proposed by Vitalik Buterin in the year 2013. It is a distributed public blockchain network that enables developers to build and deploy various types of next-generation decentralised applications easily. Ethereum helps in designing and issuing cryptocurrencies and tradeable digital tokens. We can even create our own democratic autonomous organisation using the Ethereum blockchain, which generally focuses on running the programming code for any decentralised application.

Features

1. Helps in tracking ownership of digital currencies.
2. Supports different command line tools (like built-in Go, Python, C++, Java, etc).
3. Provides the Ethereum wallet.
4. Enables developers to hold and protect their crypto-assets.
5. Simplifies smart contract development.

Advantages

1. Any third party cannot update or make any changes to the data.
2. It is corruption- and tamper-proof, making censorship impossible.

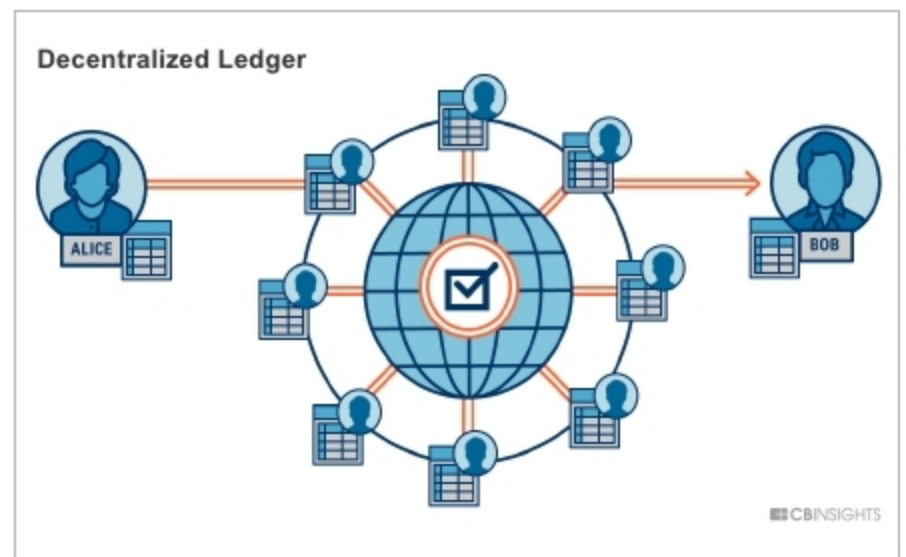


Figure 4: Architecture of a decentralised ledger system (Image source: [googleimages.com](https://www.googleimages.com))

3. Provides high security. Applications in Ethereum are well protected against any hacking attacks or fraudulent activities.
4. Zero downtime as applications never go down.

BigchainDB

This is an open source distributed ledger system that is designed for storing large data sets, and enabling its developers to deploy various blockchain Proofs-of-Concepts and applications. The important thing about BigchainDB is that it's based on a production-ready database. This makes it really appealing for deployment in production without any need to wait for the rest of the blockchain platforms to catch up. The network of BigchainDB can be public, permissioned or private, as per the access permissions that entities have over the system. In the case of a public BigchainDB, any of the participants can have access to the network or can deploy their own node and connect it to the database federation. A permissioned version can be managed by a consortium or any governing body whereby each member of the consortium is responsible for managing his or her own node in the network.

Features

1. It has the ability to deploy a decentralised system using various production-ready technologies.
2. BigchainDB is built using a federation of enterprise-ready DB nodes, like MongoDB instances, that store immutable information about assets in a synchronised manner.
3. Supports custom digital assets and also helps in establishing access permissions at a transaction level.
4. High speed transaction processing.
5. Low latency.
6. Powerful query functionality.

Advantages

1. Available as open source.
2. Highly customisable.
3. Permissions can be set at the transaction level to enforce selective access.